LUNARIS

LUNAR UNCHARTED NAVIGATION, AUTONOMY & RESOURCE INTELLIGENCE SYSTEM

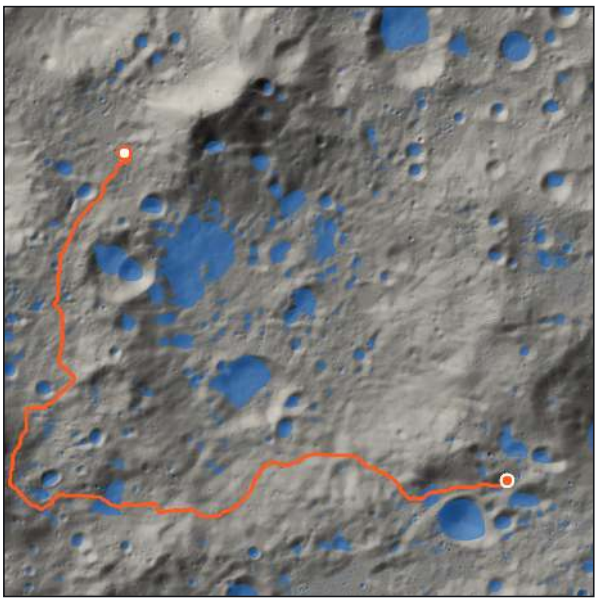
Traverse and Volatile-Prospect Assessment

Cabeus · Lunar South Polar Region

Issued 30 October 2026 · Prepared with the LUNARIS analysis pipeline

DOCUMENT CONTROL

Report number	LUN-TR-20261030-E7E27D
Date of issue	2026-10-30 09:39 UTC
Study region	Cabeus
Dataset	Cabeus crater — LOLA 80 m/px (real data)
Region centre	85.33 S, 317.87 E · D = 100.6 km
Analysis grid	943 x 943 cells @ 160 m
Configuration	0372e762659aaafc
Dataset checksum	af3fbaab303c5d09
Primary sources	NASA LRO: LOLA (PGDA), Diviner (PDS), LROC NAC; LOLA illumination models
Distribution	Unrestricted — derived from NASA open data



LOLA shaded relief, PSR (blue) and candidate traverse (orange)

EXECUTIVE SUMMARY

Summary of findings

This report documents an orbital-data assessment of the Cabeus study area using the LUNARIS pipeline. Terrain derived from LOLA laser altimetry classifies 2% of the analysis area as low mobility hazard and 38% as impassable at the adopted slope and roughness thresholds. Permanently shadowed regions occupy 8.6% of the window. A hazard-weighted candidate traverse of 261.3 km was identified, with a maximum slope of 18.2° and no impassable cells crossed. The volatile investigation-priority model flags 6.6% of the area as candidates for further investigation. A traverse simulation of the NASA VIPER (Volatiles Investigating Polar Exploration Rover) returned **FAIL** (PSR exposure 16.1 h > 9 h limit).

ROUTE LENGTH	MAX SLOPE ON ROUTE	PERMANENT SHADOW	ROVER SIMULATION
261.3 km	18.2°	8.6%	FAIL

SECTION 01

Introduction and scope

LUNARIS is a decision-support system for permanently shadowed regions (PSRs) near the lunar poles. It combines laser altimetry, thermal-model products, illumination models and orbital imagery into traceable data layers, plans rover traverses over a documented cost surface, and evaluates them with a terramechanics-based rover simulation. Every layer in this report is classified as measured, derived, model estimate or illustrative, and cells without valid data are treated as unknown rather than safe.

Study region. 151 x 151 km window centred on the IAU gazetteer centre of Cabeus (-85.33°, 317.87°E, D = 100.6 km). Named features inside the window (IAU gazetteer): Cabeus, Mons Cabeus, Cartwright, Bay (partial).

SECTION 02

Data sources

Table 1. Input data products and checksums.

Layer	Product	Instrument	Units	SHA-256
elevation	LDEM_80S_80MPP_ADJ	LRO LOLA	m, height relative to 1737.4 km reference sphere	4c130e1306ed
elevation uncertainty	LDEM_80S_80MPP_ADJ_ERR	LRO LOLA	m	dbd37f9b3c9f
effective resolution	LDEM_80S_80MPP_ADJ_EFFRES	LRO LOLA	m	2063ea22c186
slope reference	LDSM_80S_80MPP_ADJ	LRO LOLA	deg	d5445a6ca17b
psr mask	LPSR_80S_20MPP_ADJ	LRO LOLA (illumination modelling)	1 = PSR, 0 = not PSR (averaged to 80 m: PSR fraction)	459785b7e5af
illumination avg	AVGVISIB_85S_060M_201608	LRO LOLA (illumination modelling)	fraction of time the solar disc is visible	d376de9ce1b7
earth visibility avg	AVGVISIB_85S_060M_201608_EARTH	LRO LOLA (visibility modelling)	fraction of time the Earth disc is visible	667667d3c526
thermal max	DLRE_PRP_SOUTH.TAB:temp_max	LRO Diviner (DLRE)	K	6bf4ff4e5b5e
thermal avg	DLRE_PRP_SOUTH.TAB:temp_avg	LRO Diviner (DLRE)	K	af7bf0c7c0ba
ice stability depth	DLRE_PRP_SOUTH.TAB:ice_depth	LRO Diviner (DLRE)	m (deeper than 2.87 m is encoded as 999)	13799d6b85b2

Table 2. Layers not ingested for this study region.

Layer	Status
image	not provided by this dataset
hydrogen	not provided by this dataset

Citation. Barker et al. (2023) PSJ 4:183 doi:10.3847/PSJ/acf3e1; Paige et al. (2010) Science 330:479; Mazarico et al. (2011) Icarus 211:1066

SECTION 03

Methods

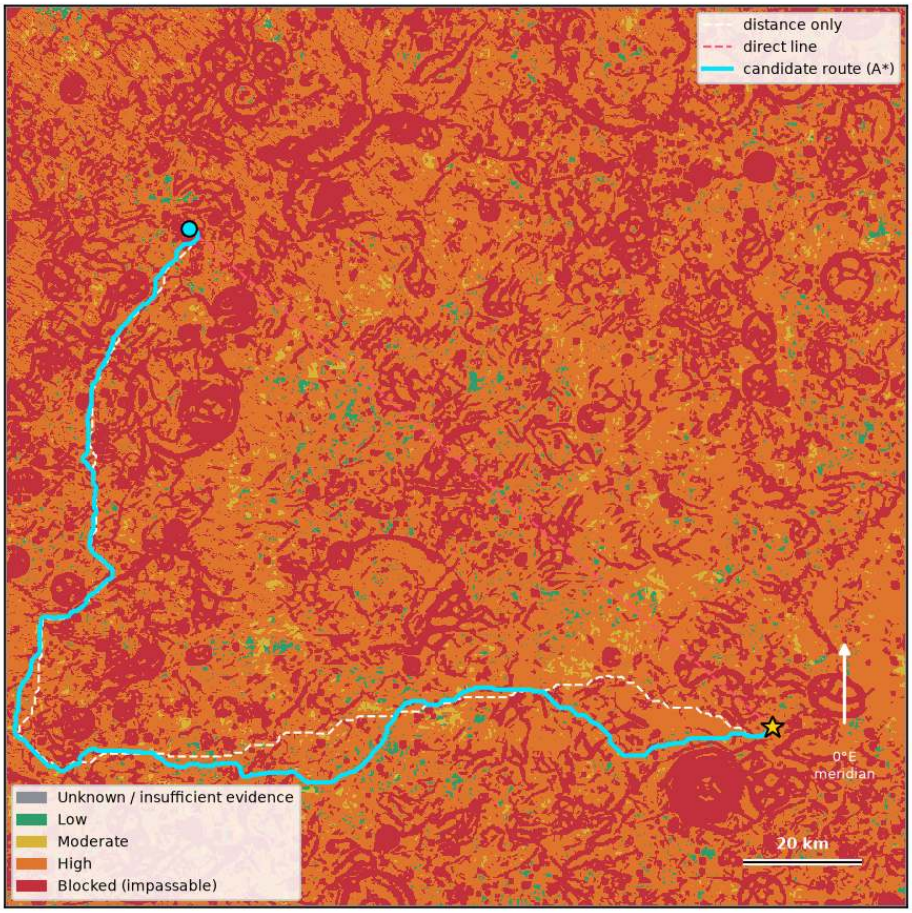
Rasters are ingested with checksum verification, aligned to the elevation grid and block-averaged to the analysis resolution (cells require at least 75% valid sub-cells). Slope uses the Horn (1981) operator; roughness is the RMS residual from a local least-squares plane. Hazard classes take the worse of the slope and roughness classes; producer uncertainty raises a cell by one class. Traverses use an 8-connected A* search with an admissible octile heuristic over a per-metre cost multiplier.

Table 3. Processing record.

Stage	Detail
Ingest	elevation — LDEM_80S_80MPP_ADJ.TIF
Ingest	elevation uncertainty — LDEM_80S_80MPP_ADJ_ERR.TIF
Ingest	effective resolution — LDEM_80S_80MPP_ADJ_EFFRES.TIF
Ingest	slope reference — LDSM_80S_80MPP_ADJ.TIF
Ingest	psr mask — LPSR_80S_20MPP_ADJ.TIF
Ingest	illumination avg — sun_avg_visibility.tif
Ingest	earth visibility avg — earth_avg_visibility.tif
Ingest	thermal max — prp_temp_max.tif
Ingest	thermal avg — prp_temp_avg.tif
Ingest	ice stability depth — prp_ice_depth.tif
Align	elevation uncertainty — identical grid (no resampling)
Align	effective resolution — identical grid (no resampling)
Align	slope reference — identical grid (no resampling)
Align	psr mask — identical grid (no resampling)
Align	illumination avg — resampled (bilinear) onto elevation grid
Align	earth visibility avg — resampled (bilinear) onto elevation grid
Align	thermal max — identical grid (no resampling)
Align	thermal avg — identical grid (no resampling)
Align	ice stability depth — identical grid (no resampling)
Resample	factor 2 — nan-aware block mean
Analyse	1.98 s · configuration 0372e762659aaafc

SECTION 04

Terrain assessment



Source: Cabeus crater — LOLA 80 m/px (real data) | Status: derived

Figure 1. Mobility hazard classes with the candidate traverse (cyan), distance-only alternative and direct line. Up is the 0°E meridian direction of the polar stereographic grid.

Table 4. Hazard class distribution over the analysis area.

Hazard class	Area fraction
Unknown / insufficient evidence	0.8%
Low	1.6%
Moderate	1.6%
High	57.9%
Blocked (impassable)	38.1%

Derived slope agrees with the producer slope map with r = 0.995 and a median absolute difference of 0.23° (a consistency check on the same source elevation model).

Boulder-scale hazards. grid is 160 m/px; rover-scale boulders (<10 m) are sub-pixel. Requires native-resolution NAC/ShadowCam imagery (~0.5–2 m/px) and a labelled training/evaluation set.

SECTION 05

Traverse planning

Table 5. Candidate traverse compared with rejected alternatives.

Metric	Candidate (hazard-weighted)	Distance-only	Direct line
Length (m)	261,343	248,674	132,195
Path cost	925,203	1,087,828	impassable
Maximum slope (°)	18.2	21.0	22.9
Mean slope (°)	7.0	8.6	9.6
Cells in PSR	56	47	114
Blocked cells crossed	0	0	231
Total ascent (m)	9,773	9,879	8,070

Path cost = sum over moves of step length x mean per-metre cost multiplier of the two cells; a value equal to the length means flat, certain, low-hazard terrain all the way. Relative units — not energy, time or risk probability.

Start 83.4104° S, 315.68° E; goal 87.5928° S, 324.73° E (planetocentric).

Table 6. Hazard exposure along the candidate traverse.

Hazard class along route	Cells
Unknown / insufficient evidence	0
Low	134
Moderate	38
High	1247
Blocked (impassable)	0

SECTION 06

Volatile investigation priority

The priority model combines psr, thermal_max, ice_stability_depth with explicit weights; layers not used: hydrogen. A cell is assessed only when thermal evidence is present. Assessed fraction: 100.0%.

Table 7. Investigation-priority class distribution.

Priority class	Area fraction
Not assessed (insufficient evidence)	0.0%
Lower relative priority	89.3%
Moderate relative priority	4.1%
Candidate for further investigation	6.6%

Under 9 weight-perturbation scenarios, 47.5% of assessed cells change class at least once; the ranking should be read with this sensitivity in mind.

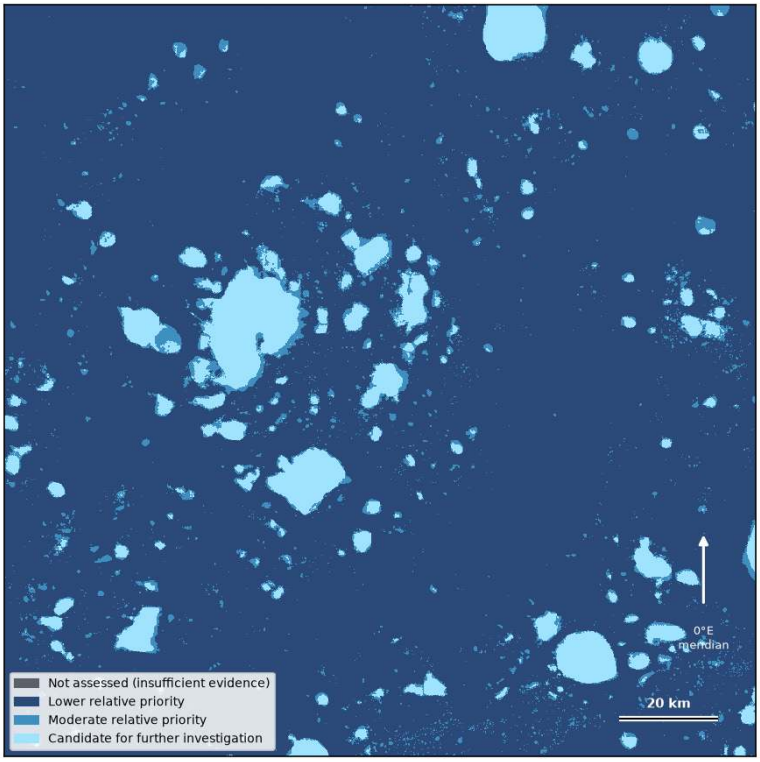


Figure 2. Relative investigation priority for water ice (ranking of evidence, not a detection).

SECTION 07

Rover traverse simulation

The NASA VIPER (Volatiles Investigating Polar Exploration Rover) was driven along the candidate traverse over the native-resolution elevation model on Lunar regolith — Apollo LRV design values, with Bekker–Wong wheel–soil forces, static stability and shadow-time checks. Verdict: **FAIL** — PSR exposure 16.1 h > 9 h limit.

Table 8. Rover simulation results.

Quantity	Value
Distance driven / route length	261.34 / 261.34 km
Drive time (continuous)	811.7 h
Maximum slope / pitch / roll	20.8° / 20.2° / 18.5°
Wheel slip, maximum / mean	65% / 10%
Maximum sinkage	23.6 mm
Minimum static stability margin	13.5°
Time in PSR (longest continuous)	32.11 h (16.05 h)
Shadow energy / usable battery	18983 / 4878 Wh
Loose-regolith sensitivity case	FAIL — immobilised: slope 16.8° needs more thrust than the soil can provide at 100 % slip (94.06 km)

Table 9. Simulation events.

Event	Distance (km)
slope 15.0° exceeds the 15° design requirement (12148 m of route)	71.82
slip 40% > 40% operational threshold (1787 m of route)	93.54
battery reserve reached while in shadow	138.54
continuous PSR exposure exceeds the 9 h limit	140.16

Unpublished rover parameters adopted as assumptions: ground clearance m, cg height m, drive efficiency, battery reserve frac. Terrain is the LOLA DEM at its native 20 m grid (bilinear). Rocks, boulders and craters smaller than ~20-40 m are invisible to this simulation; VIPER's published 15 cm obstacle limit cannot be checked from this data. Wheel-soil forces use the Bekker-Wong model with Apollo LRV soil parameters (Costes et al. 1972). Real PSR regolith properties are unmeasured and may differ; see the loose-soil sensitivity run. Quasi-static model: no dynamics, wheel-level suspension articulation, slip-sinkage or turn-in-place time.

SECTION 08

Assumptions

- Slope classes [10.0, 15.0, 25.0] deg and blocked >= 25.0 deg are prototype assumptions.
- Roughness high >= 2.0 m, blocked >= 6.0 m RMS (prototype).
- Cost weights {'slope': 8.0, 'roughness': 3.0, 'uncertainty': 4.0, 'psr': 1.5} are prototype choices; cost is relative, not energy or time.
- Ice-priority weights {'psr': 0.35, 'thermal_max': 0.4, 'ice_stability_depth': 0.25} with sensitivity +/-50%.
- Ground distance equals grid distance (polar stereographic scale error < 0.04 % within 2 deg of the pole).
- No-data and unknown cells are treated as blocked for routing.

SECTION 09

Limitations

- Analysis grid 160.0 m/px: rover-scale obstacles (boulders, small craters) are not resolved.
- PSR map, temperatures and ice-stability depth are model products, each with its own epoch and resolution.
- No LEND hydrogen layer was used; no direct measurement of water ice is included.
- Illumination, communications line-of-sight and thermal survival are not modelled in the route.
- Layer 'LEND hydrogen' unavailable: not provided by this dataset

APPENDIX A

Layer catalogue

Table 10. Analysis layers, epistemic status and method.

Layer	Status	Units	Method
Elevation (LOLA)	measured	m	Block-mean 2x2 of source DEM (cells need >=75% valid sub-pixels).
Hillshade (visual aid)	illustrative	unitless	Lambertian shading of Horn gradients.
Slope	derived	degrees	Horn (1981) finite differences on the analysis grid; NaN propagates from no-data.
Roughness (detrended RMS)	derived	m	Least-squares plane removal per window (separable correlation).
DEM uncertainty (PGDA)	derived	m	Block mean.
DEM effective resolution	derived	m	Block mean.
Elevation flagged uncertain	derived	boolean	Threshold on producer uncertainty layers (prototype thresholds).
Slope — PGDA reference	derived	degrees	Block mean of 20 m slope.
Permanently shadowed regions	derived	fraction of cell	Block mean of the producer's 0/1 PSR map (illumination modelling on LOLA topography).
Annual max temperature (Diviner PRP)	derived	K	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Annual mean temperature, 2 cm depth (Diviner PRP)	derived	K	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Ice stability depth (Diviner PRP)	derived	m	Thermal-model fit (PDS PRP) resampled nearest-facet then block-averaged.
Average sun visibility (18.6-yr)	derived	fraction of time	LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean.
Average Earth visibility (18.6-yr)	derived	fraction of time	LOLA horizon modelling, hourly over 18.6 yr; 60 m product resampled bilinearly, then block mean.
LEND hydrogen	unavailable	-	-
Mobility hazard class	derived	class	Slope edges [10.0, 15.0, 25.0] deg, blocked >= 25.0 deg; roughness high >= 2.0 m, blocked >= 6.0 m.
Traversal cost multiplier	derived	cost per metre (>= 1)	See core/navigation/cost.py
Ice investigation priority (score)	model_estimate	0–1 relative	core/ice_priority/evidence.py
Ice investigation priority (class)	model_estimate	class	Thresholds at 1/3 and 2/3.
Evidence coverage	derived	fraction of weight	-
Priority sensitivity (std)	derived	score std	One-at-a-time perturbation.

Prepared with LUNARIS (Lunar Uncharted Navigation, Autonomy & Resource Intelligence System). Configuration 0372e762659aaafc; dataset checksum af3fbaab303c5d09; generated 2026-10-30T09:39:30 UTC.